

# Operator Discovery Report: perturbation\_double\_bridge\_escalating\_2opt\_escape\_transfer

- Operator type: perturbation.
- Host scaffold: nearest\_neighbor\_2opt.
- Core idea: Deterministic stagnation escape for 2-opt-based TSP scaffolds: first apply a structured double-bridge to break nearest-neighbor traps, then fall back to segment reversals within a fixed window when repeated stagnation is
- Novelty classification: deterministic\_perturbation\_schedule.
- Rediscovery signature: perturbation:double\_bridge:segment\_reversal:3:3.
- Surviving candidate: False.
- Problem class where it helps: none confirmed.

## Pseudocode

- Track stagnation count during local search and restarts.
- Use double\_bridge by default and escalate toward segment\_reversal after the stagnation threshold.
- Increase perturbation strength deterministically when escape pressure rises.

## Complexity

- Non-empty lines: 12.
- Complexity score: 0.44.
- Fingerprint: 1db1adae4e01961c.

## Why It Should Help

- The operator targets local-minimum escape directly by controlling when and how restarts are perturbed.

## Failure Cases

- Does not transplant cleanly into clustered\_local\_search.
- No validation family shows a clear improvement over the host baseline.
- Disabling the operator does not hurt, so the ablation does not support a real operator effect.

## Transplant Results

- nearest\_neighbor\_2opt gap delta 0.0, runtime delta 46.782171, distance-eval delta 0.0.
- cheapest\_insertion\_2opt gap delta 0.0, runtime delta 1060.664743, distance-eval delta 0.0.
- random\_restart\_2opt gap delta 0.0, runtime delta 28.251957, distance-eval delta 802.285714.
- sparse\_three\_opt gap delta 0.0, runtime delta -2.922929, distance-eval delta 0.0.
- clustered\_local\_search gap delta 0.015717, runtime delta -16.912314, distance-eval delta -699.428571.

## Instance-Family Results

- heldout\_tsplib operator gap 0.235058 vs baseline 0.235058.

- clustered\_tsp operator gap 0.110853 vs baseline 0.110853.
- elongated\_corridor\_tsp operator gap 0.003204 vs baseline 0.003204.
- grid\_like\_tsp operator gap 0.106696 vs baseline 0.106696.
- nearest\_neighbor\_trap\_tsp operator gap 0.0 vs baseline 0.0.
- two\_cluster\_bottleneck\_tsp operator gap 0.109057 vs baseline 0.109057.
- uniform\_euclidean operator gap 0.0 vs baseline 0.0.

## **Pareto Results**

- Gap delta: 0.0.
- Runtime inflation: -0.002078.
- Distance-evaluation inflation: 0.0.
- Same-gap-faster flag: True.
- Better-gap-same-runtime flag: False.

## **Ablation Results**

- disabled mean gap 0.121317 and delta vs full operator 0.0.
- simplified\_perturbation mean gap 0.121317 and delta vs full operator 0.0.
- alternate\_perturbation mean gap 0.121317 and delta vs full operator 0.0.